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EXTENDED-ABSTRACT

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Exploring Visual Discomfort and Opportunities for Vision Augmentations: Visual Noise Cancellation and Head-worn LCD Light Actuators for Perception Modulation

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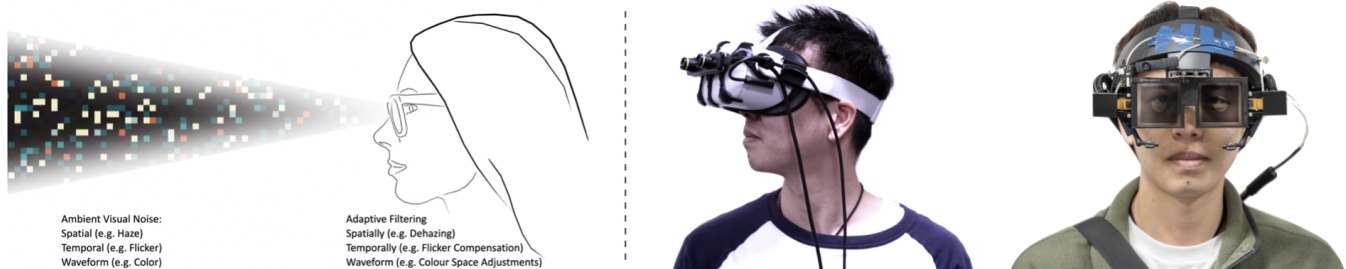


Figure 1: The concept of Visual Noise Cancellation (VNC). (Left) We characterise Visual Noise and propose active cancellation by way of computerised glasses. (Right)

ABSTRACT

Recently, Head-Mounted Displays have shown great potential for augmenting vision beyond traditional visual aids, such as compensating for complex visual impairments or providing superhuman vision to the unimpaired eye. Example applications addressed colour vision deficiency or emphasised visual cues for people with low vision. However, Visual Discomfort or Visual Noise and how to address it in vision augmentations are less explored, which is surprising given that acoustic noise control or cancellation is widely applied in modern audio headphones. In my PhD, we explore the area of Visual Noise and Discomfort through a series of studies. Specifically, we explore the general problem of what represents Visual Discomfort and Visual Noise, prototypically explore the first prototypes to receive feedback on the concept of Visual Noise cancellation, including novel prototypes for vision augmentation using LCD light actuators, also review the literature for the current state of vision augmentation.

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CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality; HCI theory, concepts and models.**

KEYWORDS

Visual Noise Cancellation, Augmented Human, Perception, Augmented Reality, Vision Augmentation, XR, Visual Discomfort, LCD actuators, Computational Glasses

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1 INTRODUCTION AND BACKGROUND

Despite the increasing popularity of VR and AR head-mounted displays (HMDs) such as Oculus Quest or the recent Apple Vision Pro, the exploration of HMDs as visual aids is only in its infancy. Recent works, including from our own group presented approaches used to address low vision [10], colour vision deficiency [9], general technical prototypes for vision augmentations [8], and even design spaces for vision augmentations [6] and some companies started

to announce first commercial products such as eSight¹, NuEyes², and IrisVision³. However, despite these first explorations within academia and industry, the full potential of vision augmentations using HMDs is not well understood and mainly focuses on low vision or colour vision deficiency.

Our work takes inspiration from audio-noise cancellation via headphones, which have become omnipresent in recent years. People do not use it only to improve the sound quality of their played music but increasingly also to separate themselves from the environmental noise, and here mainly human-made noises. Our initial research questions are what forms Visual Noise if it is an actual problem and what opportunities exist to address it using vision augmentations with HMD technology traditionally used for AR or VR [3]. So far, Visual Noise or Visual Discomfort is only rarely addressed in the academic literature. Usually defined as something visually unpleasant [7] but not caused by visual impairments, there has never been an actual definition or discussion of what encompasses Visual Noise or Discomfort. Some recent studies interpret Visual Noise as a distracting movement in the periphery and propose to use HMDs to blur the surroundings to reduce the distractions [4]. However, a more holistic approach to understanding and addressing Visual Noise and Discomfort is missing. Based on this, the main research question exploring visual discomfort and opportunities for vision augmentation can be divided into three subsets:

- What are the existing approaches for vision augmentation, their research methods, and findings?
- What is Visual Noise, and what is the general perception of the concept of Visual Noise cancellation?
- What are technical solutions for compensating for Visual Noise using vision augmentation techniques and HMDs?

2 METHODS AND CURRENT STATUS

Reflecting on the different characters of our research questions, we present in the following our approach and methods for exploring our research questions but also address the status of our research.

2.1 Survey on Vision Augmentations

We start by brainstorming key information that we want to extract when surveying the field of vision augmentations. Identified information include the used techniques and devices (e.g. video-see through, optical see-through, etc.), application area(s), user study details including the inclusion of target users to only name a few of the parameters we are interested. We then started to look into known papers addressing vision augmentations, such as the well-known works on Foresee[10] and Cuesee[11] and our own work and extracted all relevant citations (including papers citing those works). Overall, this led to about 50 related papers being added to the list. We then went through this list of related papers and applied the same strategy of extracting and citing related papers. This led to around 200 overall papers. We have finished the reading tasks and have started to systematically extract the key information for further analysis in our planned paper.

¹<https://www.esighteyewear.com/>

²<https://www.nueyes.com/>

³<https://irisvision.com/>

2.2 Visual Noise and Visual Noise Cancellation

Since the initial discussion among around 15 HCI-related researchers, we noticed that different from defining or measuring audio noise with its frequency or amplitude, Visual Noise is more subjective and complex to give an accurate and logical definition under rigorous mathematical science. We took inspiration from speculative design space[1] so that we could capture as many Visual Noise scenarios as possible. Following IDEO[5] brainstorming rules and steps, we run four workshops with participants from different backgrounds. As part of the discussion in those workshops, participants must: 1) brainstorm what Visual Noise is and give as many Visual Noise scenarios as possible; 2) brainstorm Visual Noise categories and categorise the former Visual Noise scenarios. Based on the result from the workshop, we then ran a converged discussion among the authors to refine ideas. Different from the Visual Noise category brainstorming in the workshop, here we try to build a Visual Noise space with category as the retrieval clue. The main challenge is that the workshop categories were repetitive, not search-oriented, and could not create a Visual Noise space well. As such, we revisited the initial inspiration from the audio noise cancellation and human visual system. Similar to acoustic noise cancellation but more in line with the visual domain, we gave three categories of source stimulus or frequency domain and two categories in Visual Impact as we saw differences in the impact of Visual Noise on individuals. We also ran an explorative investigation with our first Visual Noise Cancellation HMD prototype based on Oculus Quest 2 and Zed mini as its external camera and typical scenarios from the workshop to collect the initial feedback on Visual Noise Cancellation Headphones. The result shows that even participants had different ideas in the workshop but were still saturated with the term Visual Noise. We also noticed the general potential of our Visual Noise Cancellation Headphones as the participants from the explorative user study showed great interest, although they responded to compensate for Visual Noise in different ways[2].

2.3 Wearable Visual Noise Cancellation using LCD Light Actuators

Most approaches for vision augmentation use devices borrowed from AR/VR that usually place a display in front of the user's eye, which are normally over-functional as those devices are entertainment-based. Meanwhile, our first exploration of Visual Noise Cancellation Headphones inspired us to know that Visual Noise Cancellation often requires us to actually remove a distracting stimulus instead of adding new information in the screen layers. Thus, we started to explore technologies that block light from entering the human eye. Specifically, we explored LCD displays in a wearable form factor. Different from current VR/AR HMDs, our LCD displays have no refractions; they directly put two LCD panels in front of the eyes, making them unfocused and low resolution. However, this character also brings several advantages to them, such as there are no light-emitting elements attached to the glasses, which makes them light and has low power consumption. While also based on this speciality and the former experience in Visual Noise, we brainstorm the potential application of LCD glasses: 1) in the spatial domain, the LCD can draw some specific patterns, and those patterns would bring beneficial effects

on the human visual system, like dark background and transparent centre part to block periphery Visual Noise; 2) in the temporal domain, although the frequency is limited by the LCD glasses, we can still use it to block headlights moving in the night; 3) in intensity, if the LCD dark level or colour can be controlled, it could potentially be used as adjustable sunglasses that the block level can adapt to the sun brightness.

We ran a qualitative and quantitative mixed-user study with three scenarios to collect the initial feedback using head-worn LCD light actuators for perception modulation. To examine the glasses' performance, we take the pupil diameter and gaze position as quantitative measurements, together with a Likert scale and open interview as qualitative and subjective feedback. We have finished the user study and just started analysing the data.

3 EXPECTED CONTRIBUTIONS

Corresponding to the research questions, our main expected contributions are as follows: 1) we will present a holistic overview of vision augmentation HMDs and discuss their limitation and potential future inspiration; 2) we converged a first Visual Noise space for defining Visual Discomfort and run an exploratory investigation with the prototype of Visual Noise cancellation headphones; 3) we explored the potential visual perception modulation applications in LCD light actuators glasses with a qualitative and quantitative mixed user study.

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